

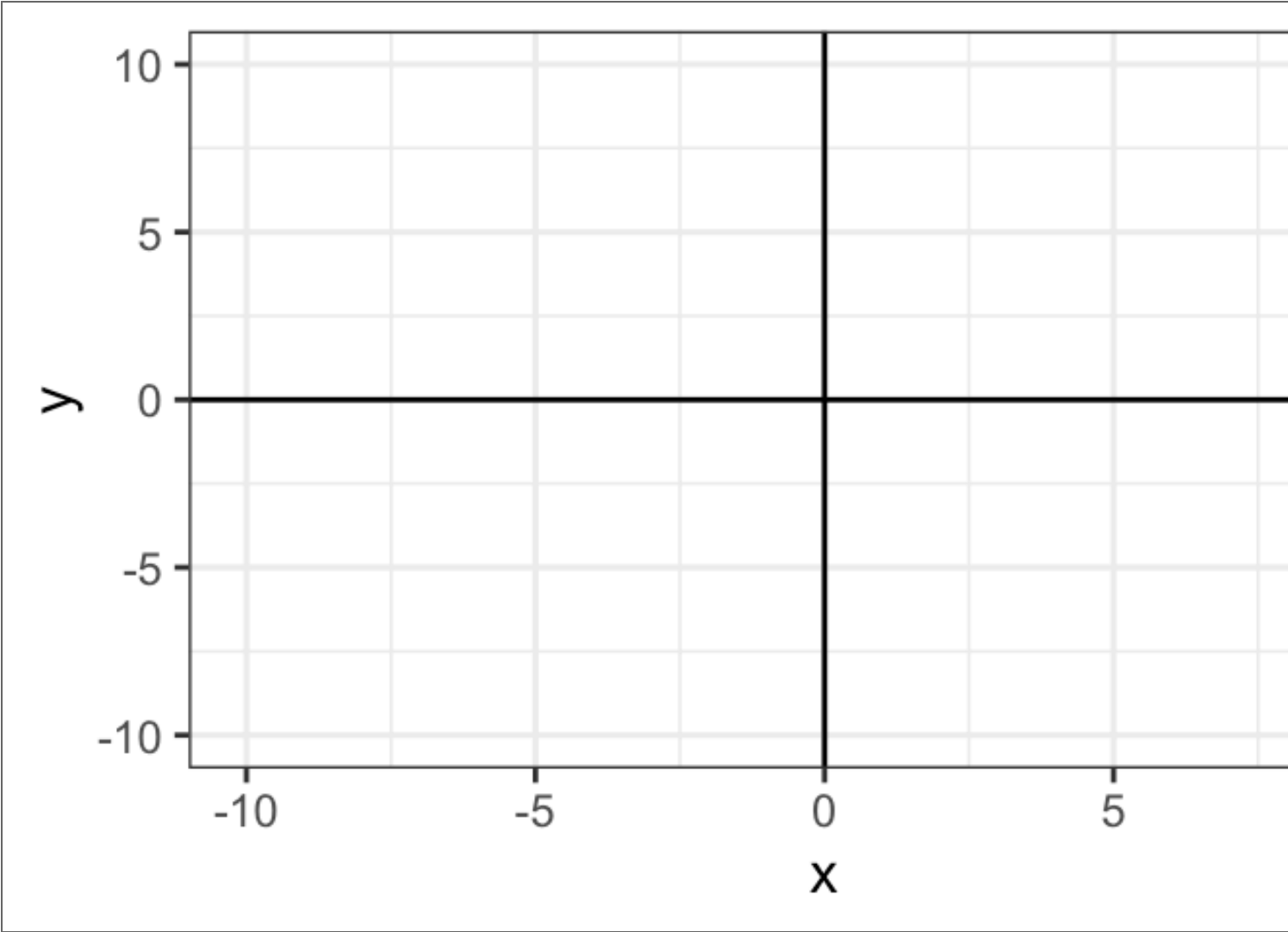
2: Basic math for statistics

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Welcome back: math time!

Coordinates and lines

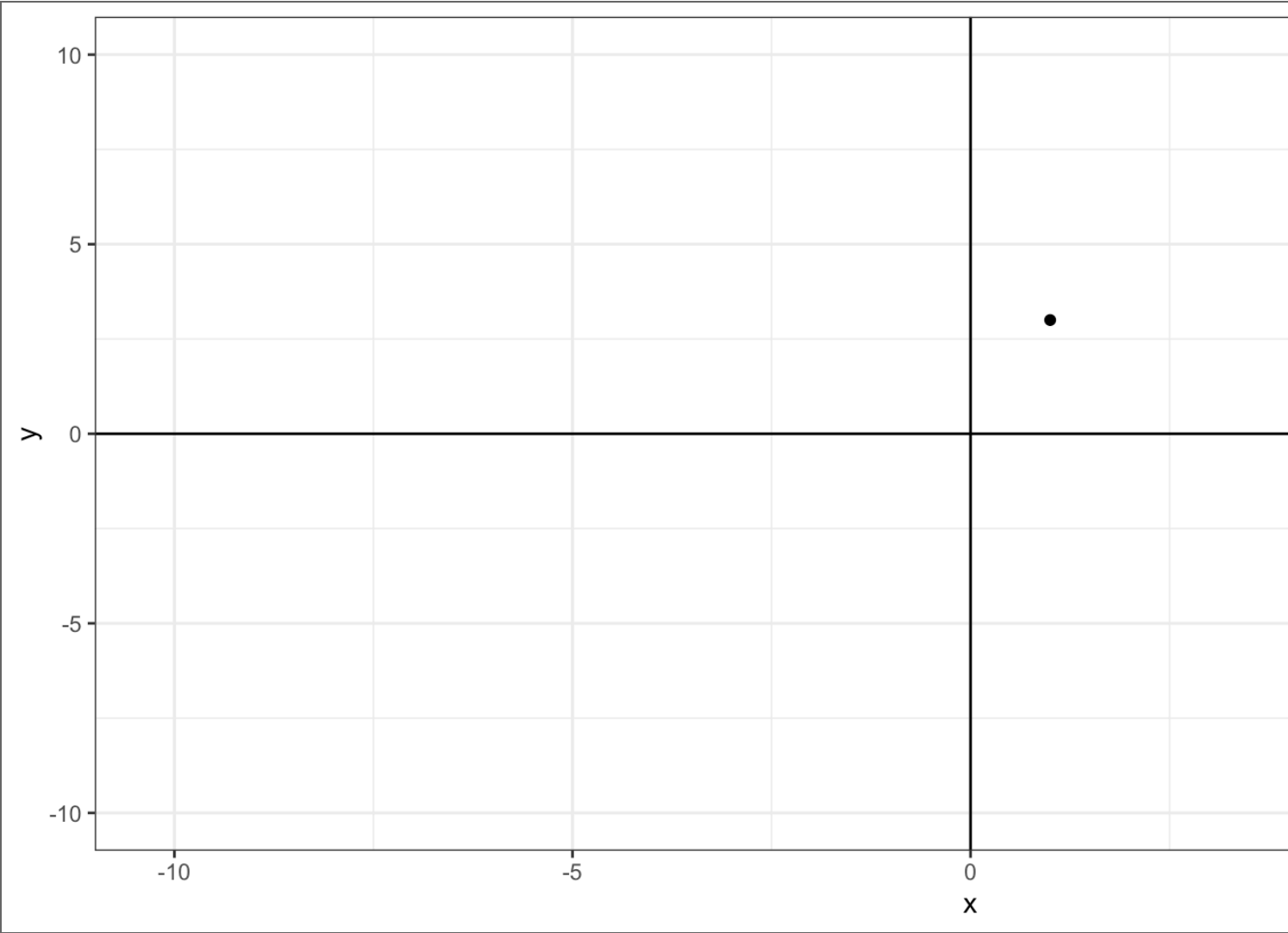
The coordinate plane



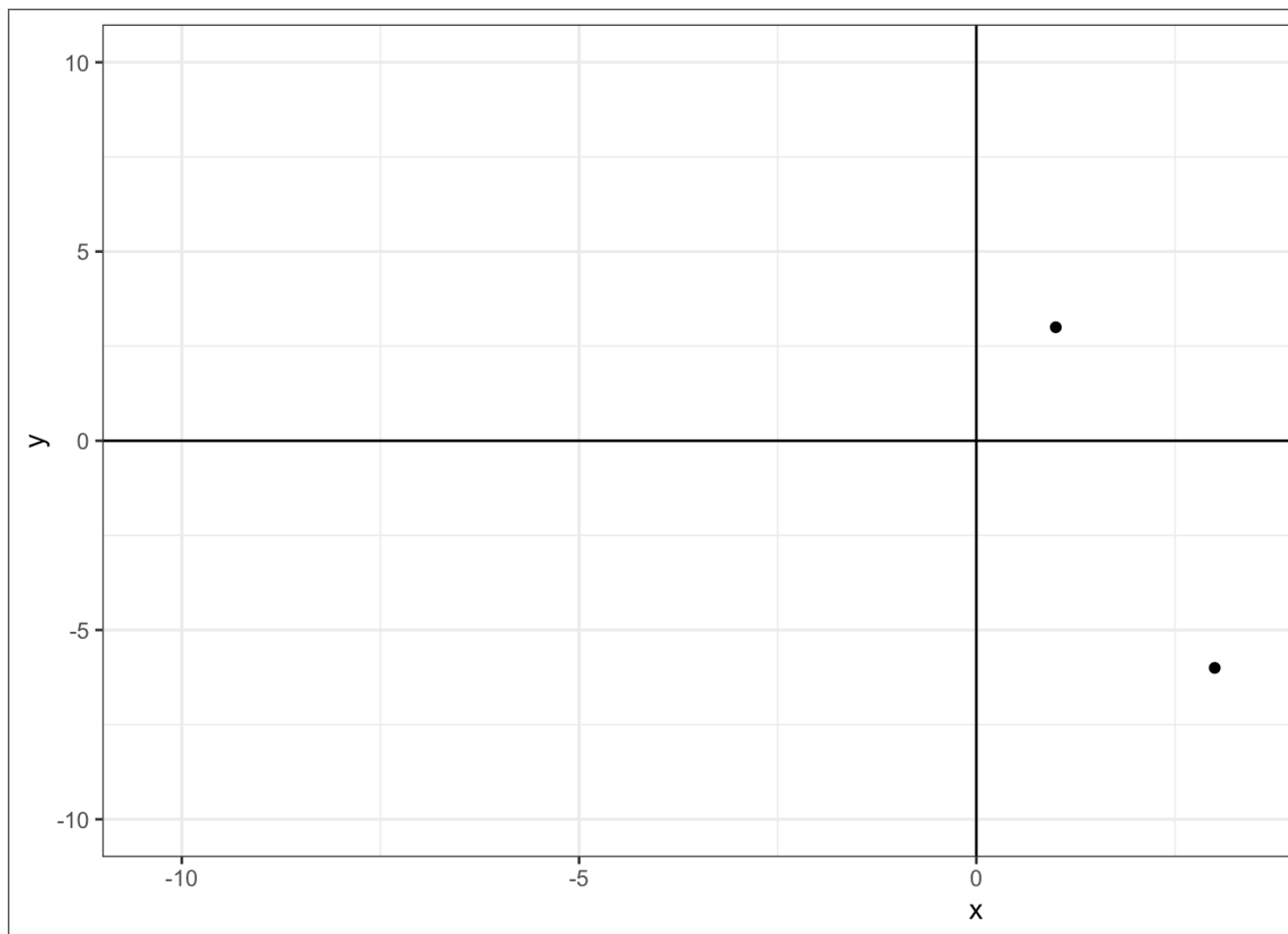
Plotting points

For coordinate pair (x_1, y_1) , we can plot along an x and y axis.

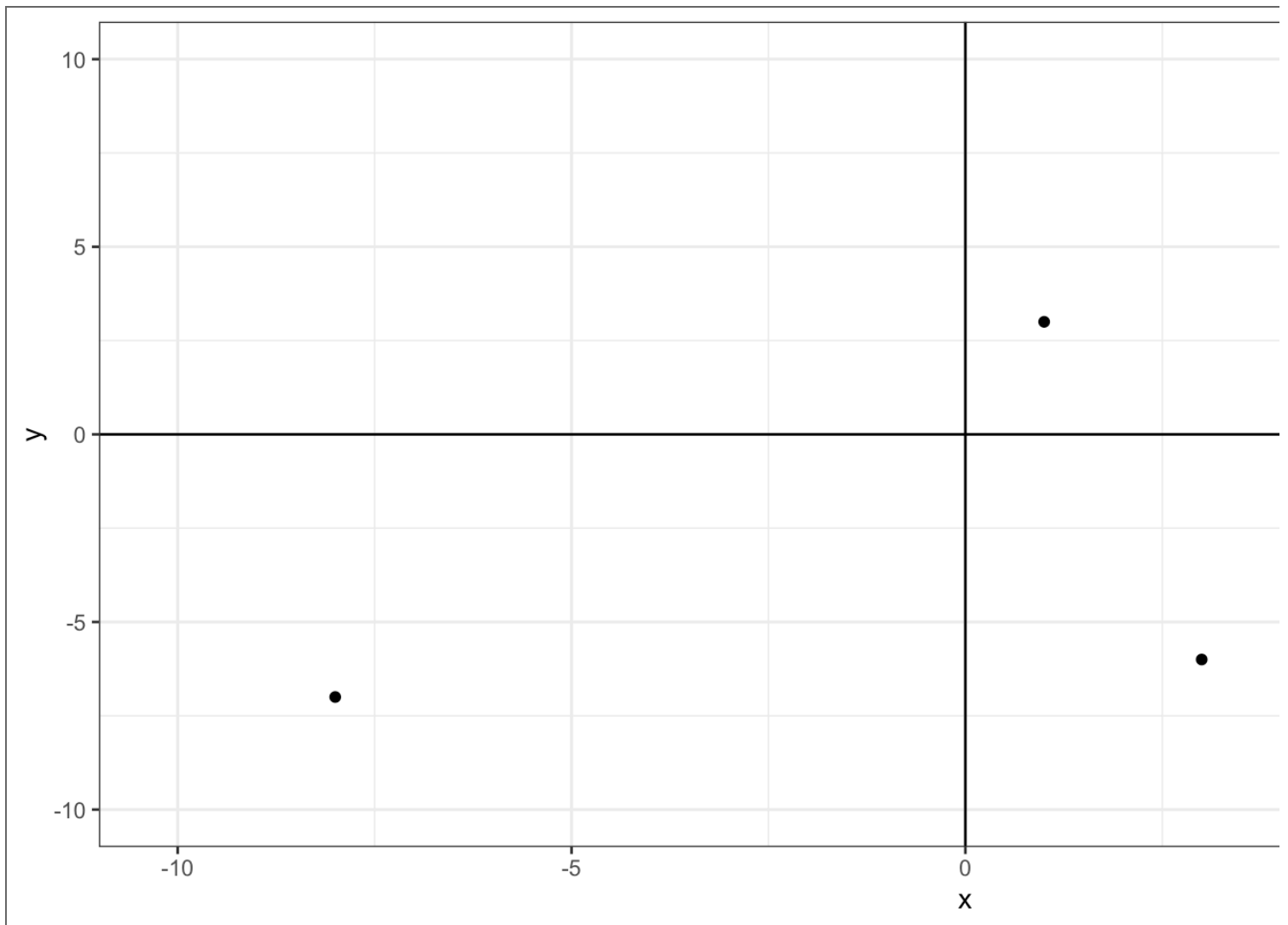
Example: (1,3)



Example: (1,3), (3,-6)



Example: (1,3), (3,-6), (-8, -7)



Practice

Sketch out a coordinate plane on paper. Plot the following points.

Remember we give points as (x, y) , where x represents the horizontal location and y represents the vertical location

- $(0, 1)$
- $(-3, 2)$
- $(5, -6)$
- $(5, 5)$

Lines

The typical equation for a line is $y = mx + b$ where m is the slope and b is the y-intercept.

You will often see a line expressed as a regression equation:

$$y = \beta_0 + \beta_1 x$$

where β_0 is the y-intercept and β_1 is the slope.

The slope

Slope measures the steepness of a line. A line with a positive slope has increasing values of y as x increases. A line with a negative slope has decreasing values of y for increasing values of x .

We can calculate the slope with two coordinates on the line $(x_1, y_1), (x_2, y_2)$

The slope is the ratio of the difference in y values to the difference in x values.

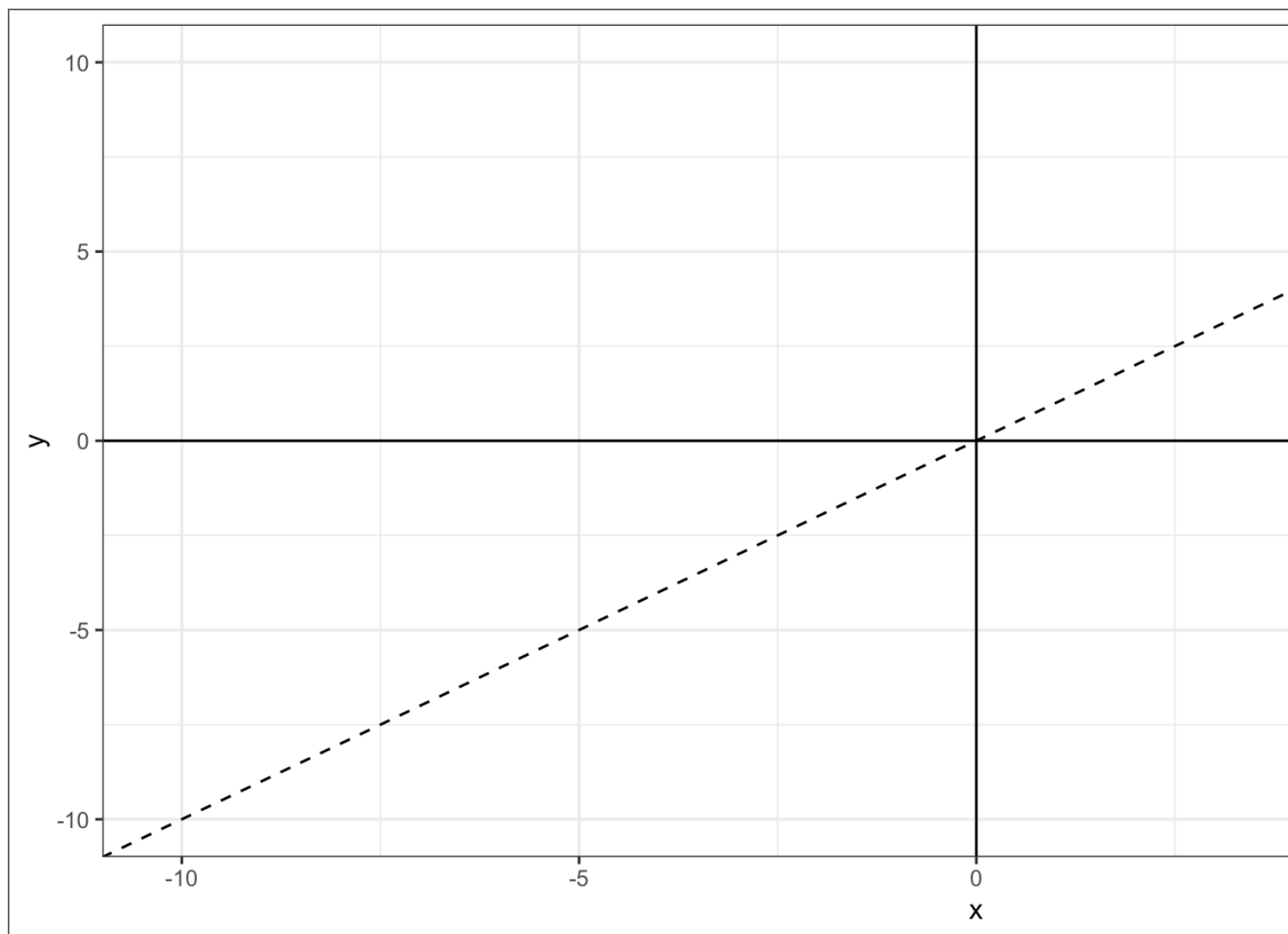
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

The y-intercept

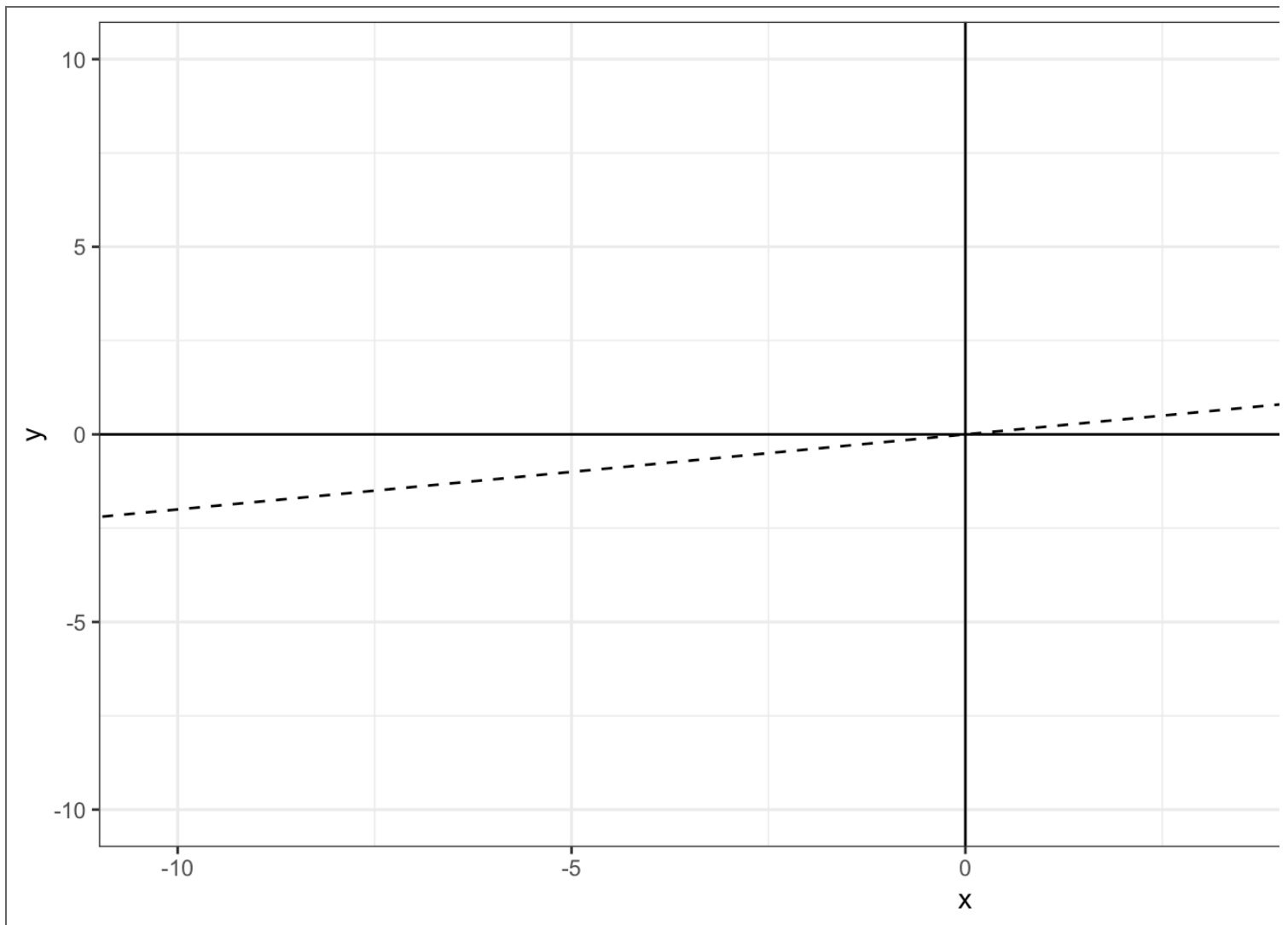
The y-intercept is the value of y when $x = 0$. If we have the value of one point on the line, and the slope, we can obtain the y-intercept

$$b = y_1 - m \cdot x_1$$

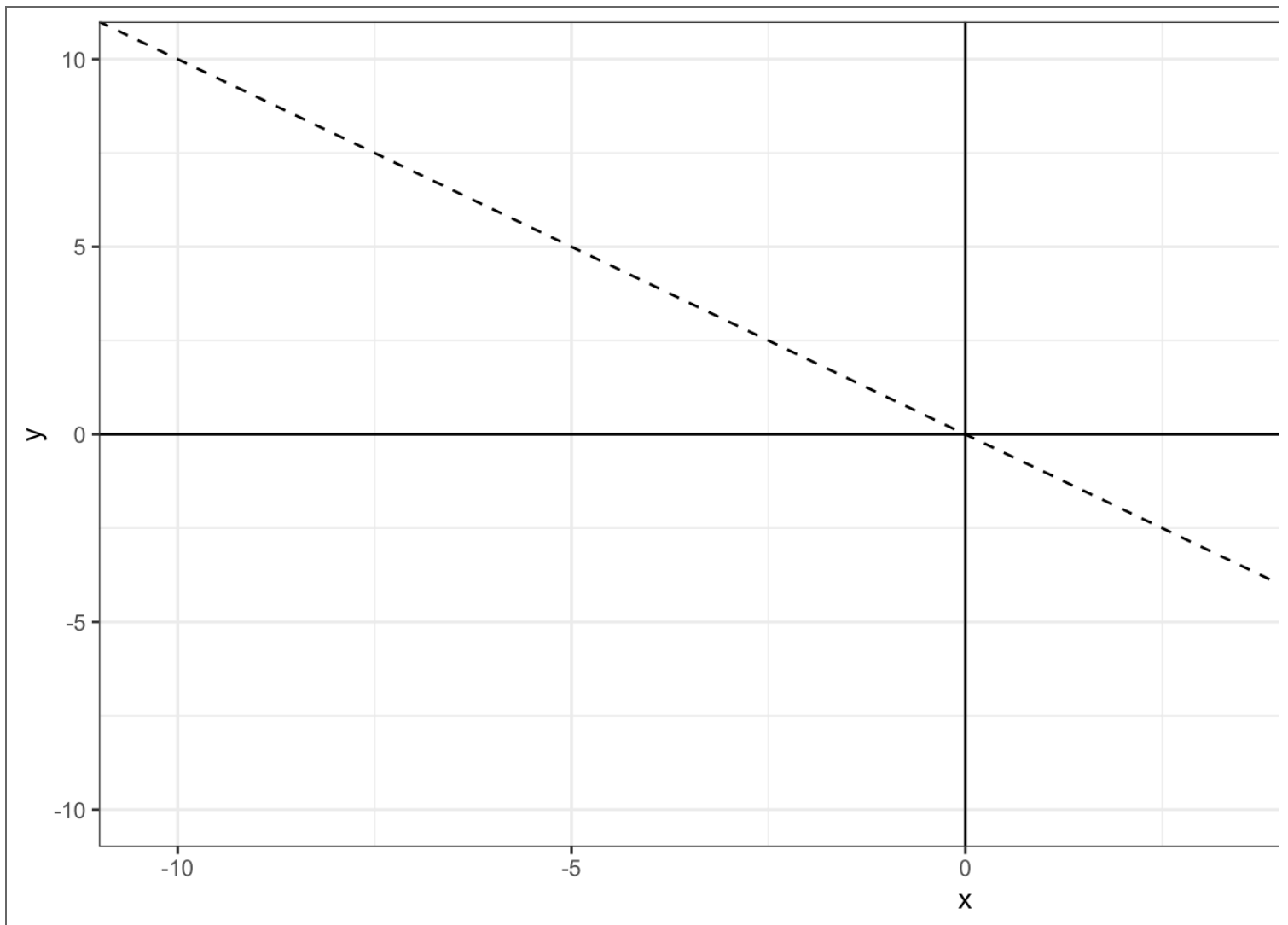
Example: intercept = 0, slope = 1



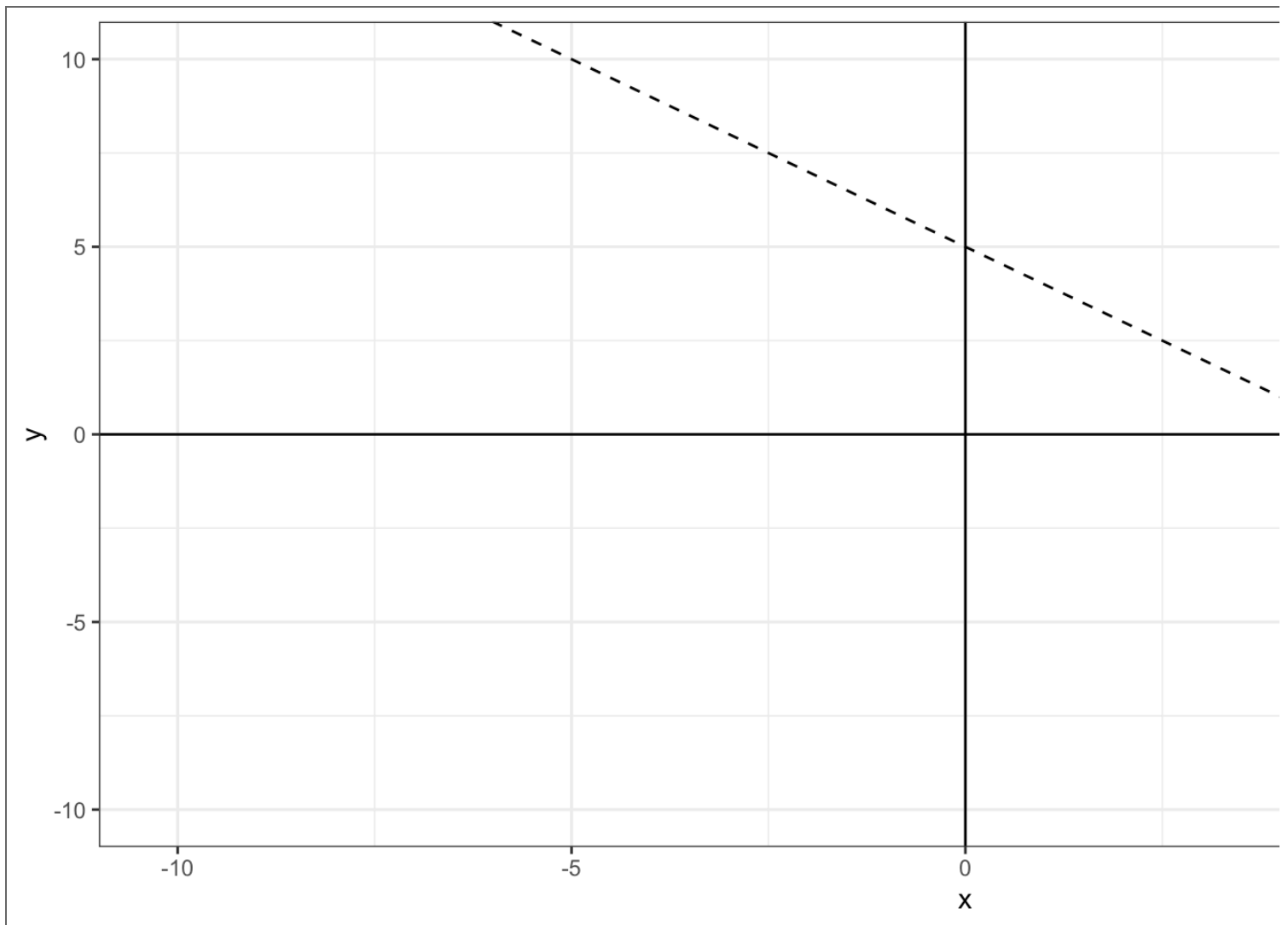
Example: intercept = 0, slope = 0.2



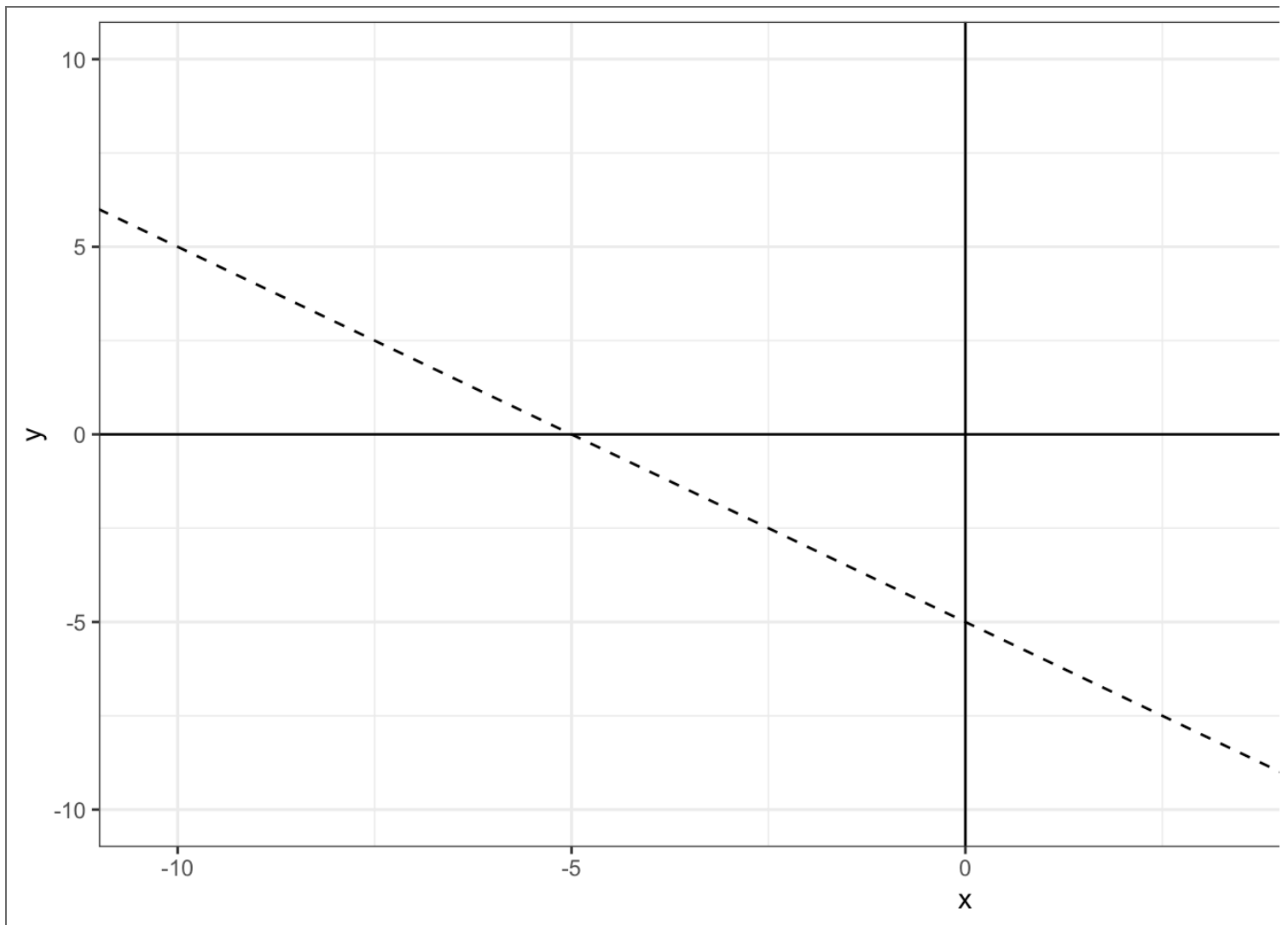
Example: intercept = 0, slope = -1



Example: intercept = 5, slope = -1



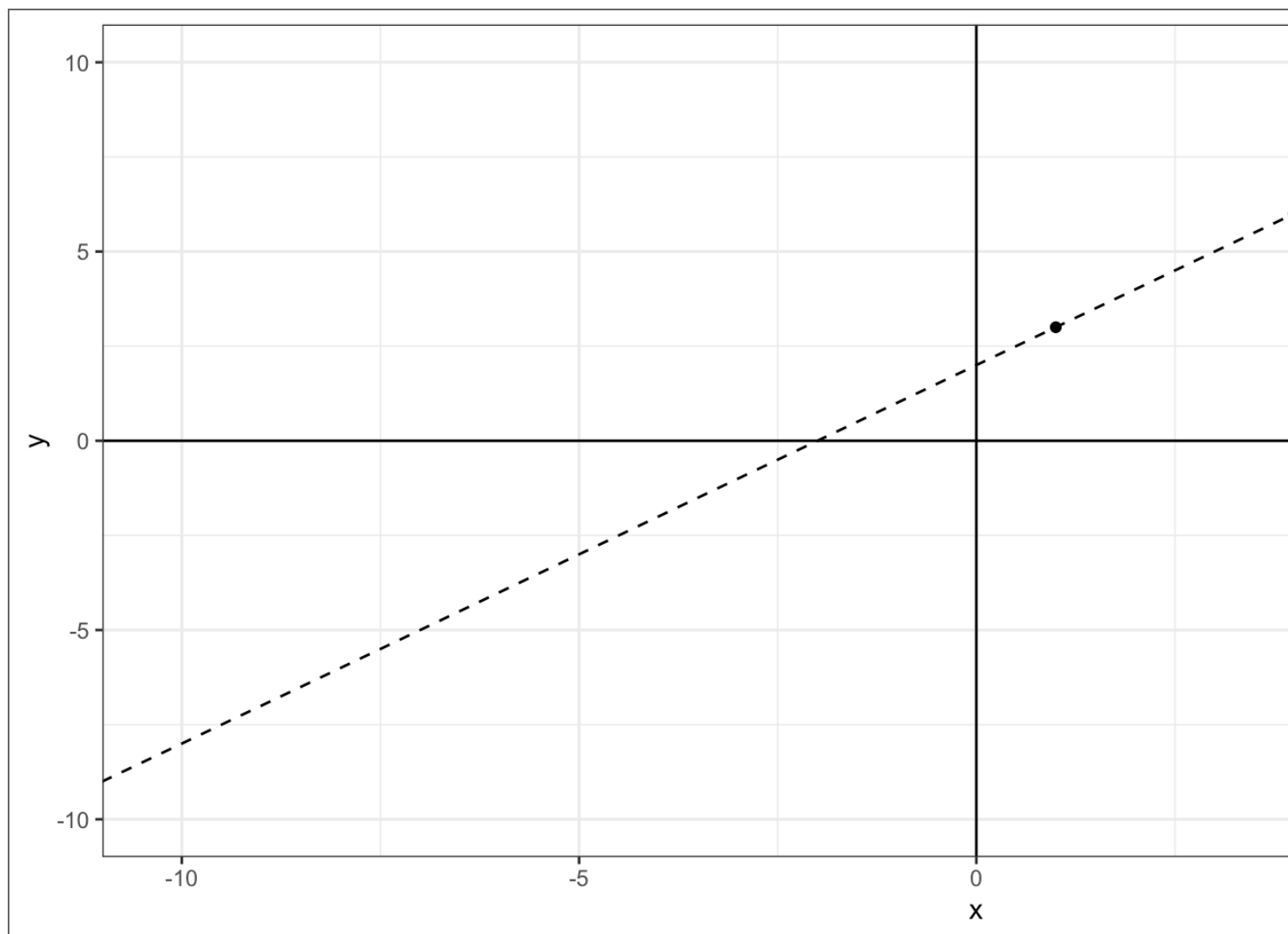
Example: intercept = -5, slope = -1



Example

- Given the points $(1, 3)$ and $(4, 6)$, compute the slope and y intercept for a line that intersects both points.
- the slope is $m = \frac{6-3}{4-1} = 1$
- the y-intercept is $b = 3 - 1 \cdot 1 = 2$. The equation of the line is $y = 1x + 2$

Example: $y = 1x + 5$



Exercises:

Compute slope, y-intercept, and sketch a line for each pair of coordinates

The equation for a line is $y = mx + b$

The slope is computed as $m = \frac{y_2 - y_1}{x_2 - x_1}$

The y intercept is $b = y_1 - m \cdot x_1$

- . $(2, 3), (5, 6)$
- . $(-2, 4), (0, 2)$
- . $(10, 12), (-5, -2)$

Functions

Functions, in general

A function maps each element in a set X to an element in set Y

- Linear function: $f(x) = x + 5$
- Quadratic function: $f(x) = x^2 + 2x + 3$
- Exponential function: $f(x) = e^{2x} + 6$

Graphical forms of functions: linear

$$f(x) = 2x + 5$$

Quadratic equations

$$f(x) = x^2 + 2x + 3$$

Exponential growth

$$f(x) = e^{2x}$$

Exponential decay

$$f(x) = 0.8^x$$

Graphical forms of functions: logarithmic

$$f(x) = \log(x)$$

Exercises

Evaluate these functions for $x = 1$, $x = 2.5$, and $x = -6$

- $f(x) = 2x$

- $f(x) = \frac{x}{2}$

- $f(x) = 2(x + 1)^3$

Summation

Represented as $\sum$, with begin and end points

$$\sum_{x=1}^3 x$$

Summation

$$\sum_{x=1}^3 x = 1 + 2 + 3 = 6$$

Summation in R

In R, we can calculate a sum using the `sum()` function

```
1 # make an integer vector from 1 to 3
2 x<-1:3
3 # x<-c(1,2,3) is equivalent
4 sum(x)
```

```
[1] 6
```

Summation exercises

Compute the following by hand, and then in R

$$\sum_{x=3}^8 (x + 1)$$

$$\sum_{x=1}^4 2x$$

Exercises (solutions)

Compute the following in R

$$\sum_{x=3}^8 (x + 1)$$

```
1 x<-3:8 # or c(3, 4, 5, 6, 7, 8)
2 sum(x+1)
```

$$\sum_{x=1}^4 2x$$

```
1 x<-c(1, 2, 3, 4) # or 1:4
2 sum(2*x)
```

Vectors

Vectors are one-dimensional arrays of values. They have dimension n , where n is the number of elements in the vector

$$x = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$$

Indexing

We can define each element with a position index i .

$$x = [3 \quad 1 \quad 5 \quad 2]$$

- What is x_2 ?
- What is x_4 ?

Vector operations

We can compute operations on vectors using single values, called scalars. Simply perform the operation on each element of the vector

$$x = \begin{bmatrix} 3 & 1 & 5 & 2 \end{bmatrix}$$

$$2 + x = \begin{bmatrix} 5 & 3 & 7 & 4 \end{bmatrix}$$

$$2x = \begin{bmatrix} 6 & 2 & 10 & 4 \end{bmatrix}$$

Vector operations

We can compute math operations on two vectors of equal dimension by performing the operation on each element pair by index

$$x = \begin{bmatrix} 3 & 1 & 5 & 2 \end{bmatrix}$$

$$y = \begin{bmatrix} 4 & -2 & 3 & 0 \end{bmatrix}$$

$$x + y = \begin{bmatrix} 7 & -1 & 8 & 2 \end{bmatrix}$$

Exercises: compute the following

$$x = \begin{bmatrix} 2 & 1 & 4 & -2 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 & 5 & 3 & 8 \end{bmatrix}$$

. $x + 2$

. $2y$

. $x + y$

. $2x - y$

Matrices

A matrix is a rectangular array of numbers, with dimensions expressed as rows $\times$ columns

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

A is a 3 x 3 matrix

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

B is a 2 x 3 matrix

$$\mathbf{C} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

C is a 3 x 2 matrix

Matrix notation

$\mathbf{X}_{ij}$ refers to the value in the i th row and j th column of matrix $\mathbf{X}$.

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix}$$

Matrix notation

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

What is

- $X_{1,2}$
- $X_{2,1}$
- $X_{1,3}$

Diagonal Matrix

A diagonal matrix has zero values except on the diagonal:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Identity Matrix

An identity matrix is a special case of a diagonal matrix, where all values on the diagonal are equal to 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Diagonal and Identity matrices are also symmetric, where all $x_{ij} = x_{ji}$. Symmetric matrices are square.

Matrix operations

Matrix transpose

A transpose interchanges the rows and columns of a matrix, or rotates it. The dimensions are switched, so an $n \times k$ matrix becomes a $k \times n$ matrix. We denote a transpose with a T

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \mathbf{B}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Addition and subtraction

Two matrices (or vectors) can be added or subtracted only if they have identical dimensions. Then add or subtract the corresponding elements of the matrices.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 + 5 & 2 + 6 \\ 3 + 7 & 4 + 8 \end{bmatrix}$$

Multiplication by scalar

Matrices and vectors can be multiplied by constant values (called scalars).

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad 4\mathbf{B} = \begin{bmatrix} 4 & 8 & 12 \\ 16 & 20 & 24 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \quad 3\mathbf{C} = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}$$

Matrices and vectors in R

The c() function makes a vector

We can define a vector in R using the `c()` function

```
1 x <- c(3, 5, 8)
2 x
```

```
[1] 3 5 8
```

Vector indexing in R

We can call a vector index i using brackets

```
1 x
```

```
[1] 3 5 8
```

```
1 x[1]
```

```
[1] 3
```

```
1 x[3]
```

```
[1] 8
```

Operations on vectors

We can also easily perform scalar operations on vectors.

Try the following

- $x + 2$
- $2x$
- $(x + 2)^2$

Further vector operations

In $\mathbb{R}$, define $x = [1, 4, 6, 9]$ and $y = [0, -2, 5, 7]$

Compute:

- $x + y$
- $2x - y$
- $(2x + 2) \times (y - 3)$

Making a matrix

```
1 Z<-matrix(data = c(1, 2, 4, 5, 6, 7, 8, 9),  
2           nrow = 4, ncol = 2)  
3  
4 Z
```

	[,1]	[,2]
[1,]	1	6
[2,]	2	7
[3,]	4	8
[4,]	5	9

- What is $Z_{2,2}$?
- What is $Z_{4,1}$?

Matrices are made up of vectors

Bracket notation in R gets us thinking in vectors.

- Try this code `Z[,1]`. What does that return? What kind of object is it?
- What do you think you will get if you run `3 + Z[,2]`

Exercises

1 Z

	[,1]	[,2]
[1,]	1	6
[2,]	2	7
[3,]	4	8
[4,]	5	9

- How could you retrieve the 3rd row of Z in R?
- Multiply the 1st column of R by 2

Speaker notes